UNIVERSITI TEKNOLOGI MALAYSIA

SEMESTER I 2014/2015

FINAL EXAM ANSWER SCHEMA

CODE OF SUBJECT	**: SCSJ2013**
NAME OF SUBJECT	**: DATA STRUCTURES AND ALGORITHMS**
YEAR / COURSE	**: 2**
TIME	**: 2 HOURS**
DATE	**:**
VENUE	**:**

INSTRUCTIONS TO THE STUDENTS:

This test book consists of **TWO** (2) parts. The layout is as follows:

PART A – 3 Questions

PART B – 2 Questions

Answer **ALL Questions** in PART A (3 questions) and select **ONLY ONE** (1) Question from Part B. Altogether you need to **answer 4 questions**.

PART A

Question 1 **[12 marks]**

a) Name the suitable searching technique. Give the reason to justify your answer.

Technique - Sequential search. (1m)
The data lists is small and sequential search can be performed on unsorted data by comparing the search key with each elements of the array with the complexity of O(n). (1m)

[2 marks]

b) What should be done to the **DATA** array to reduce the complexity time of the searching process? Explain your answer.

Answer: Sort the array so that the number of elements that need to be traversed in the searching process can be reduced. For searching on ascending list, the search key will be compared one by one until the searched key is found. Or until the searched key value is smaller than the item compared in the list. If this situation happen the searching process stopped. This will minimize the searching process.

[2 marks]

c) Perform binary search for searching **key 20**. Show the tracing of your search.

[3 marks]

left	right	middle	NUM[middle]	found
0	8	4	9	False
5	8	6	13	False
7	8	7	17	False
8	8	8	19	False
9	8			

Search has reached the last element in the search area, with left > right. Search is terminated, and search key is not found.

d) Perform binary search for searching **key 5** . Show the tracing of your search.

[3 marks]

left	right	middle	NUM[middle]	found
0	8	4	9	False
0	3	1	3	False
2	3	2	5	True

e) Discuss the differences between the number of steps and the complexity time when searching for **key equals to 19** on **NUM** array using binary search and sequential search technique.

Answer:
Searching for search key 19 using `Binary_Search()` function requires 4 steps while searching using `Sequential_Search()` function requires 9 steps.
The complexity time for searching using `Binary_Search()` function is $O(\log_2 n)$ while the complexity time searching using `Sequential_Search()` function is O(n).

[2 marks]

Question 2 **[19 marks]**

a) A queue can be implemented using circular array or circular linked list data structure. Give one advantage of queue using circular array and one advantage of queue using circular linked list. What kind of problem that both implementation can solve.

[2 marks]

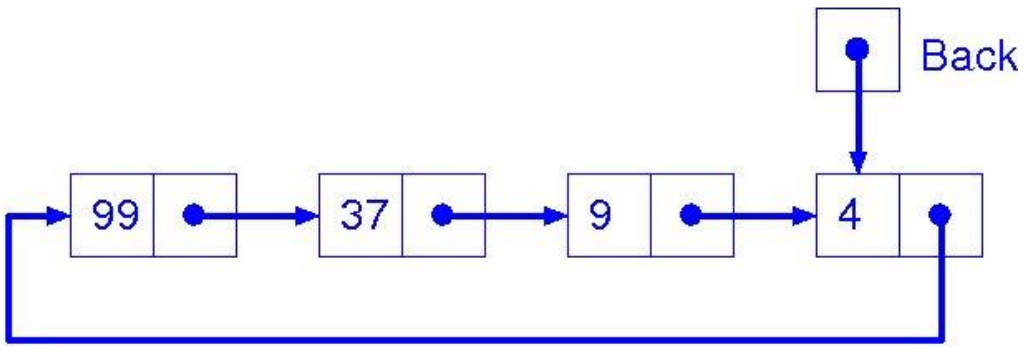
Advantage of queue circular array -> no rightward drift, queue will be full when all space of the array fully utilized. (0.5m)
Advantage of queue circular linked list -> no wasted space, since memory will be created dynamically. Only need one pointer to access the first and last node of the queue. (0.5m)
Problem solved – Memory usage issue, reduced wasted space (1m)

b) Give the differences between the implementation of the queue implementation array and queue. [3 marks]

OPERATIONS	**Queue implementation array**	**Queue implementation circular array**
Queue declaration (1m)	No variable count	Need to add variable count
`isEmpty()` function. (1m)	Back < rear	Count == 0
`isFull()` function. (1m)	Rear == maxSize - 1	Count == maxSize

c) Redraw the queue in Figure 4 that shows the current content of the queue.

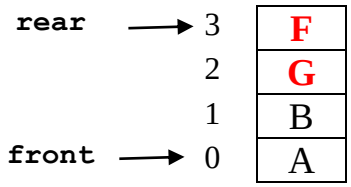
[3 marks]



1M – number of nodes = 4
1M - correct sequence and value of link list items
1M – correct position of Back pointer

d) Perform the following tasks to Queue 1:

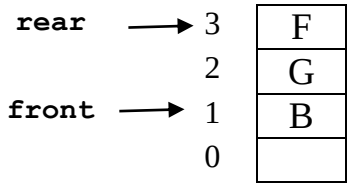
i. Redraw Queue 1 after `enQueue(G)` followed by `enQueue(F)`. Label the correct location of the `front` and `rear`.



(1m)

ii. Redraw the queue in question (d- i) after `deQueue()` and `enQueue(G)`. Label the correct location of the `front` and `rear`.

[3 marks]



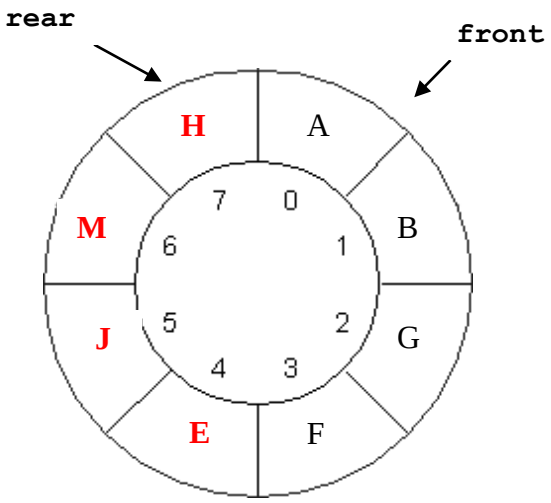
(2m)

Cannot insert G to the queue, queue already full.

e)

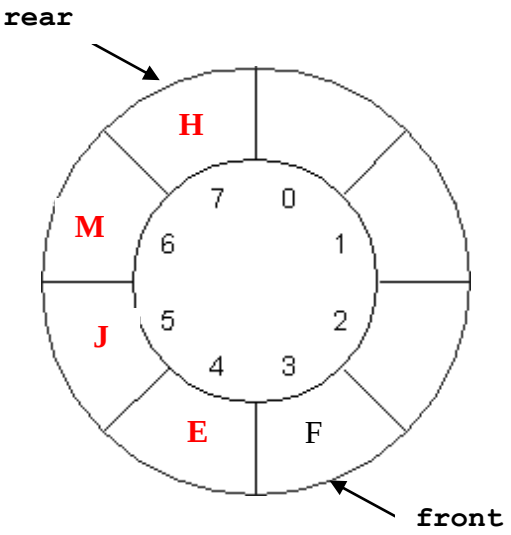
i. Redraw Queue 2 after **E, J, M, H** are **enQueue** in sequence. Label the correct location of the `front` and `rear`.

[2 marks]



ii. Redraw updated queue in question (i) after **3 deQueue** operations are performed . Label the correct location of the **`front`** and **`rear`**.

[2 marks]

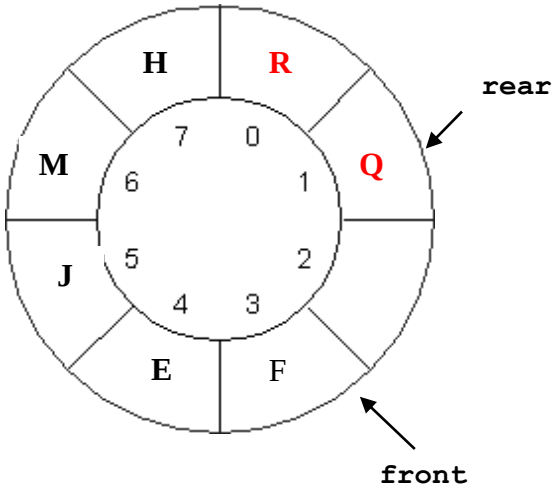


iii. List all the deQueue items in question (ii) in FIFO sequence.

A B G [1.5 marks]

iv. Redraw updated queue in (ii) after **R and Q** are **enQueue** in sequence. Label the correct location of the **`front`** and **`rear`**.

[1 marks]



v. After a series of **enQueue** and **deQueue** operations, compare the efficiency of memory usage for Queue1 and Queue2. [1.5 marks]

Rightward drift occur in Queue 1, whereby no more item can be inserted even though there are empty space in the queue, thus waste memory (1m)

Queue 2 – no wasted memory, since rear and back can circle to the front index of the array, if there is empty space in the queue. (0.5m)

Question 3 **[19 marks]**

a) Discuss the advantage of full tree and complete tree in terms of complexity related to insertion, deletion and searching operations. [2 marks]

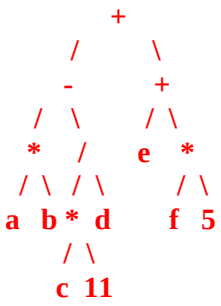
Full tree and complete tree are balanced tree with the most efficient time for searching, deletion and insertion. The complexity is O(log n).

b) Discuss the drawbacks of Binary Search Tree in terms of complexity related to insertion, deletion and searching operations. [2 marks]

Drawbacks of Binary Search Tree – the shape of binary search tree depends on the order of the sequence of key inserted to the tree and may caused unbalanced tree. If the tree is not balances the complexity related to insertion, deletion and searching operations is O(n) which is similar to linked list complexity.

c) Draw the expression tree for the following arithmetic expression. [3 marks]

a * b – (c * 11 / d) + (e + f * 5)



d) Give the inorder, preorder and postorder traversal of the tree in Figure 7. [6 marks]

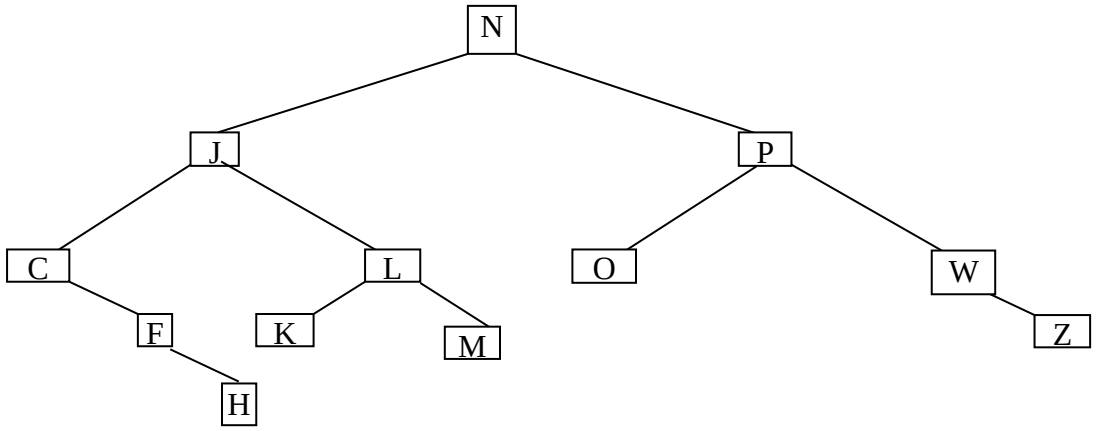
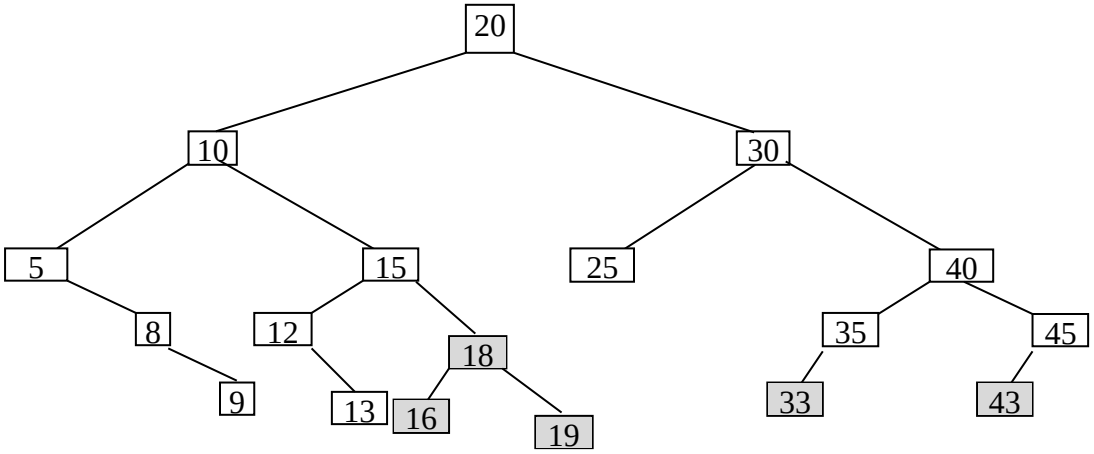


Figure 7: Binary Search Tree of char value

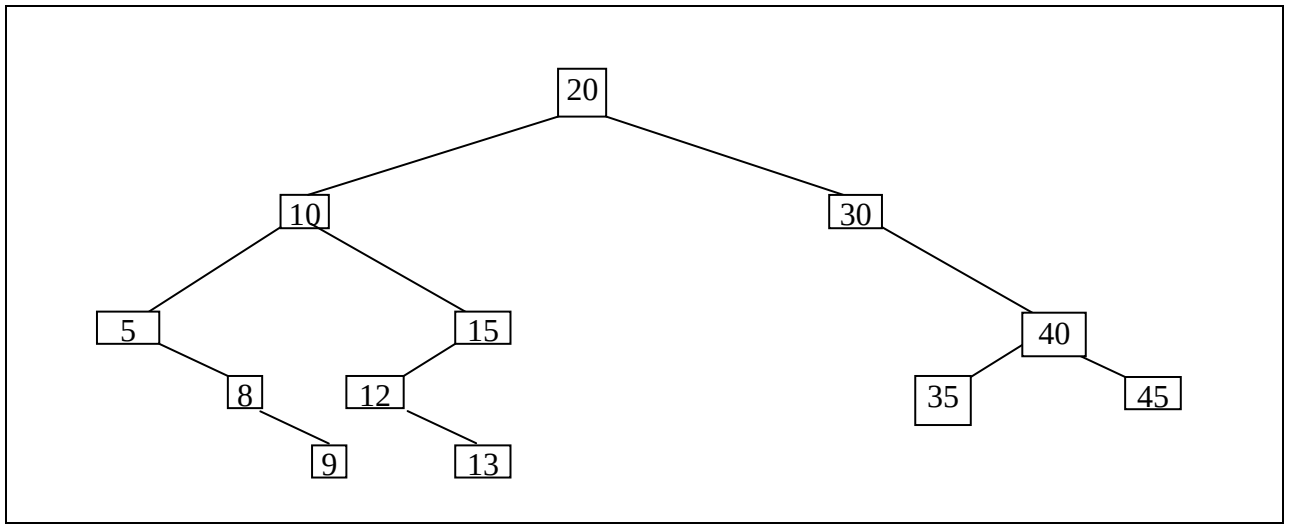
the inorder traversal – **C F H J K L M N O P W Z**
the preorder traversal – **N J C F H L K M P O W Z**
the postorder traversal – **H F C K M L J O Z W P N**

e) Redraw the tree in Figure 8 after nodes with values **18, 16, 19, 33 and 43** are inserted in sequence. [2 marks]

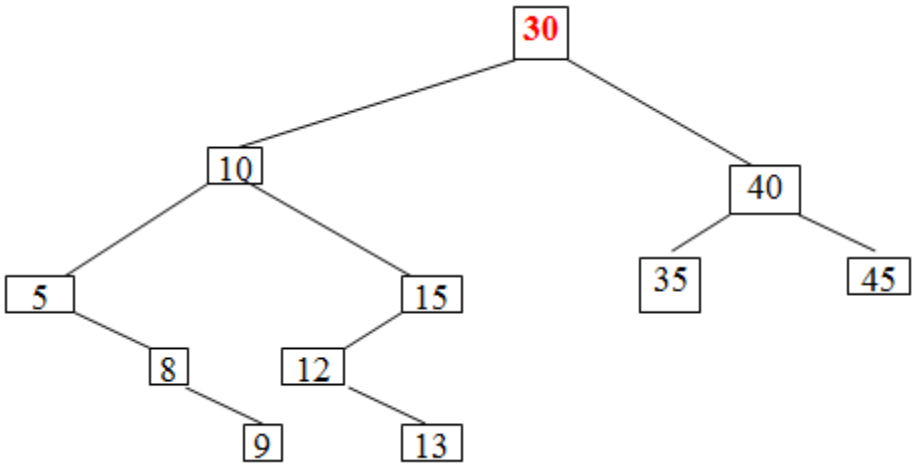


f) Redraw the tree in Figure 8 after the following nodes are deleted **in sequence**. Show the new tree after every deletion.

i) Delete 25 (1m)

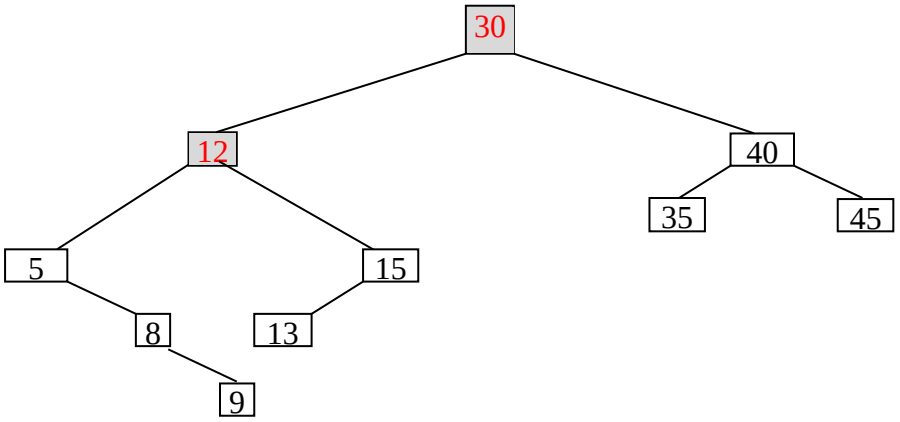


ii) Delete 20 (1.5 m)



iii) Delete 10

[1.5 m]



Note:

Deletion approach for node with 2 children in this schema – take the minimum element from right subtree.

Can also consider:

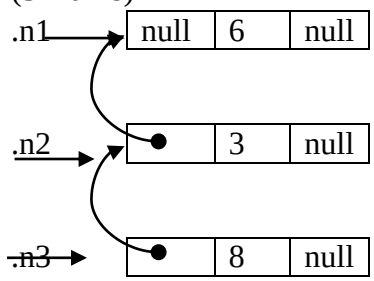
Another possible answer, to delete node with 2 children– take the maximum element from left subtree

PART B

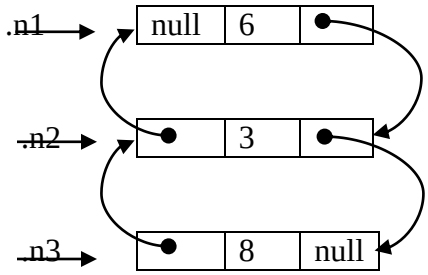
SELECT AND ANSWER ONLY ONE (1) QUESTION IN THIS PART.

Question 1 **[20 marks]**

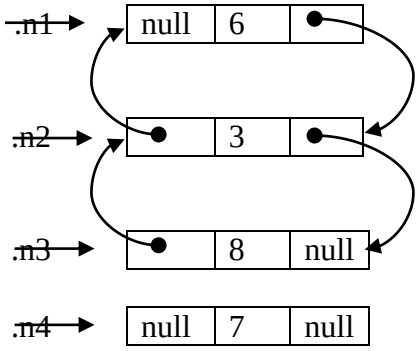
a. (3 marks)



b. (3 marks)



c. (3 marks)



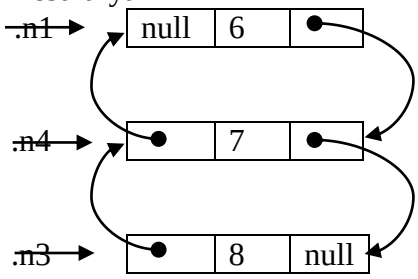
d. (3 marks)

```
n4->prev=n2->prev;
n4->next=n2->next;
delete n2;
```

(ini juga diterima)

```
NODE *del_ptr;
del_ptr=n2;
n4->prev=n2->prev;
n4->next=n2->next;
delete del_ptr;
```

-resultnya-



e. (3 marks)

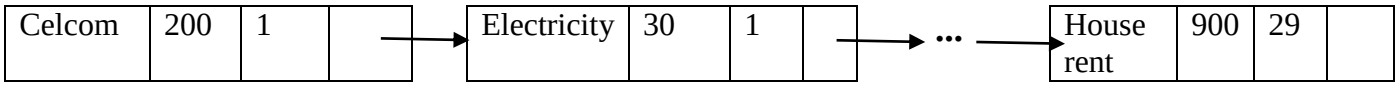
~~NODE *print_ptr;~~
~~Print_ptr = n1;~~

```
cout<<print_ptr->item;
print_ptr=print_ptr->next;
cout<<print_ptr->item;
print_ptr=print_ptr->next;
cout<<print_ptr->item;
print_ptr=print_ptr->next;
```

Atau

```
while(print_ptr!=NULL){
        cout<<print_ptr->item;
        print_ptr=print_ptr->next;
}
```

f) Possible answer: array, linked list Or circular linked list (Any answer (1m)
Justify Items (bills info) can be stored in a sorted list (array or pointer) by due date. Every nodes contain important information regarding the bill. (2pt) Rajag (2pt)



Question 2 **[20 marks]**

i) Show the content of the stack after each implementation of the sources codes based on the situation below. Assume the current content of the stack before statement Line 5 is executed as shown in first stack in the Table 2 . [3 marks]

[4]		Good Luck	Good Luck				
[3]		Luck	Luck	Luck		Bye!	
[2]	**Good**	**Good**	**Good**	**Good**	**Good**	**Good**	**Good**
[1]	**Bye!**	**Bye!**	**Bye!**	**Bye!**	**Bye!**	**Bye!**	**Bye!**
[0]	**Thank You**	**Thank You**	**Thank You**	**Thank You**	**Thank You**	**Thank You**	**Thank You**
		After Line 5-8	After Line 9	After Line 11	t After Line 12	After Line 13	After Line 15

ii) Output Displayed: **Good Bye!**

Good Luck [2 mark]

b) Using stack operations, show step-by-step conversion of the following infix expression to postfix notation. **A – B * (C + D * E) / F** [5 marks]

Infix	Stack	Postfix
A – B * (C + D * E) / F		A
– B * (C + D * E) / F	-	
B * (C + D * E) / F		AB
*** (C + D * E) / F**	-*	AB
(C + D * E) / F	-*(	AB
C + D * E) / F	-*(	ABC
+ D * E) / F	-*(+	ABC
D * E) / F	-*(+	ABCD
*** E) / F**	-*(+*	ABCD
E) / F	-*(+*	ABCDE
) / F	-*	ABCDE*+
/ F	-/	ABCDE*+*
F	-/	ABCDE*+*F
		ABCDE*+*F/-

c) Evaluate the following postfix expression using stack concept: **3 6 – 10 + 5 ***

[4 marks]

Postfix	Ch	Opr	Opn1	Opn2	Result	Stack
3 6 – 10 + 5 *	3					3
6 – 10 + 5 *	6					36
-10 + 5 *	-	-	6	3	-3	-3
10 + 5 *	10					-3 10
+ 5 *	+	+	10	-3	7	7
5 *	5					7 5
*****	*	*	5	7	35	35

d) i)What is the most suitable data structure that the compiler should use in keeping track of returned addresses and parameters when executing the program.

Stack – applying concept LIFO [1 mark]

ii) Draw a diagram to show the content of the data structure for each step of execution (steps i – iv) in illustrating the process. Based on the diagram, provide description of the operation on the data structure.

[5 mark]

Step i: main function calls function A, push the data

Return address	Parameters/arguments
1054	*a, b, c*

Step ii: function B is called, push the data

Returned address	Parameters/arguments
304	*a, b, d, e*
1054	*a, b, c*

Step iii: function C is called, push the data

Return address	Parameters/arguments
154	*a, d, e, f*
304	*a, b, d, e*
1054	*a, b, c*

Step iv: function C returns to function B and execute statement at address 154 (next pop data)

Return address	Parameters/arguments
154	*a, d, e, f*
304	*a, b, d, e*
1054	*a, b, c*

Step v: function B returns to function A and execute statement at address 304 (next pop data)

Return address	Parameters/arguments
304	*a, b, d, e*
54	*a, b, c*

Step vi: function A returns to main and execute statement at address 54 (next pop the data, the stack is empty)

Return address	Parameters/arguments
54	*a, b, c*